

EPS 101
Field Geology and Digital Mapping
Final Report



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Geological Evolution of the East Bay Hills Region: Tectonic plate interactions at a continental margin driving basin formation, magmatism and uplift

ABSTRACT

This study aims to explore the geological evolution of the Berkeley Hills, situated along the eastern margin of San Francisco Bay within California's Coast Ranges. Through comprehensive fieldwork and analysis, we examined Middle to Late Miocene stratigraphic units, including the marine Claremont Formation, terrestrial Orinda Formation, and volcanic Moraga Formation, siesta Formation and Bald Peak Basalt. Our findings reveal a complex interplay of tectonic processes, sedimentation, and volcanism associated with the transition from a convergent to a transform margin during the evolution of the San Andreas Fault system. The Claremont Formation's deep marine deposits indicate significant subsidence, while the overlying Orinda Formation's fluvial sediments suggest uplift and erosion linked to the Mendocino Triple Junction's northward migration. The Moraga Formation's basaltic flows, particularly evident in the Sibley Volcanic Regional Preserve, reflect slab-gap volcanism during this tectonic reorganization. Structural analyses, including the identification of the Siesta Valley syncline and activity along the Hayward Fault, underscore the region's dynamic deformation history.

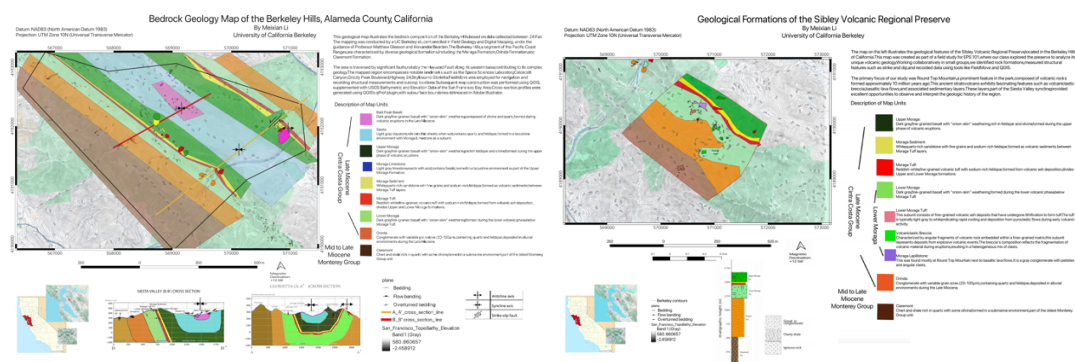


Figure 1: *Bedrock Geology Map of the Berkeley Hills and Sibley Volcanic Regional Preserve.*

INTRODUCTION

Regional Setting

The Berkeley Hills, located along the eastern margin of San Francisco Bay, are a prominent feature of California's Coast Ranges. This region records a complex geologic history tied to its tectonic setting along the evolving San Andreas Fault system. During the transition from a convergent margin to a transform margin, interactions between the Pacific and North

American Plates led to basin formation, magmatism, and significant structural deformation. These tectonic processes have profoundly influenced the stratigraphic framework and landscape of the Berkeley Hills.

The area preserves Middle to Late Miocene stratigraphic units, transitioning from marine sediments in the Claremont Formation of the Monterey Group to terrestrial deposits in the Orinda Formation and volcanics in the Moraga Formation (Gerasimov et al., 2023). These units provide valuable insights into the region's evolution during the Miocene, including uplift and subsidence events associated with the passage of the Mendocino Triple Junction and slab-gap volcanism.

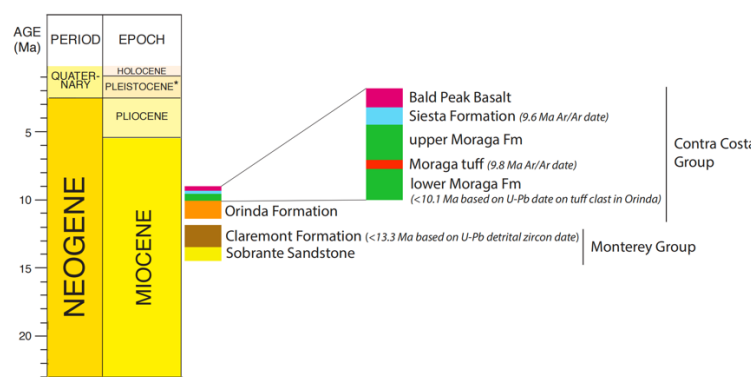


Figure 2: Geological map of the Berkeley Hills, illustrating the distribution of Middle to Late Miocene stratigraphic units,

Geography and Exposure

Fieldwork was conducted in the Berkeley Hills, specifically encompassing Grizzly Peak Boulevard, Round Top Mountain, and the Sibley Volcanic Regional Preserve. The mapped area spans approximately **10 square kilometers**, with exposures of key stratigraphic units including the Claremont Formation, Orinda Formation, Moraga Formation, Siesta Formation, and Bald Peak basalt flows. These areas are accessible and preservation of distinct geologic features, providing critical data for stratigraphic and structural analysis. It highlights the variability in lithology and depositional environments within the Middle to Late Miocene succession. Exposures in the Siesta Valley syncline reveal fine-grained cherts and shales of the Claremont Formation transitioning upward into fluvial sandstones and conglomerates of the Orinda Formation. Overlying these units are basaltic and dacitic flows of the Moraga Formation, showcasing the interplay of sedimentation and volcanism during tectonic reorganization (Graham et al., 1984)

Base Maps

The base maps for this study were developed using Google Terrain Maps, which provide detailed topographic information including hillshade and contour lines.

The coordinate system employed is NAD83 (North American Datum 1983) within UTM Zone 10N, suitable for the study area in California. To delineate the study area's boundaries, a vector shapefile was created, outlining the specific region of interest.

Contour intervals of 10 meters were utilized, with every fifth contour (50 meters) emphasized to enhance readability. The declination for the area was determined to be approximately 13.5° East as of the current date, and all compass-clinometers were adjusted accordingly to ensure accurate field measurements.



Figure 3: *Locality of the Berkeley Hills, California, and the Hayward Faults*

METHODS

The geological mapping for this study was conducted using a combination of field-based data collection and digital mapping tools. Below are the methods employed during the mapping process:

Field Methods

FieldMove Software: Fieldnotes, photos, outcrops were recorded in the field using the FieldMove app on tablets. This software allowed for real-time georeferenced mapping, capturing key structural and lithological observations directly on digital

maps.

Strike and Dip Measurements: Structural measurements, including strike and dip of bedding planes and fault surfaces, were taken using the FieldMove software (mainly) and compasses..

Field Equipment: Essential field tools included:

Geological Hammer: For sampling and closer examination of rock features.

Hand Lens: For examining fine-grained textures and mineral compositions.

Scale Bar and Ruler: Used for visualizing measurements and capturing the scale in photographs.

Data Visualization and Map Production

QGIS: The collected field data were imported into QGIS for visualization and spatial analysis. Georeferenced field points, structural measurements, and observations were digitized to create a detailed geological map of the study area.

Adobe Photoshop (PS) and Illustrator (AI): The final map figures were refined using Photoshop and Illustrator for enhanced presentation quality. These tools allowed for clear annotation, color-coding of units, and addition of legends, scale bars, and labels.

Data

Project Kml file and Plane CSV: data collected from the field.

Google Road map: used as a base map.

Berkeley contour map: identify the height

STRATIGRAPHY AND INTERPRETED

PALEOENVIRONMENTS

Claremont Formation (Middle to Late Miocene)

The Claremont Formation, part of the Monterey Group, represents the basal unit of the studied sequence in the Berkeley Hills. It is characterized by interbedded siliceous cherts, porcelanites, and shales, with depositional environments interpreted as deep marine. These sediments were deposited in a bathyal setting (~500–1,500 m depth), reflecting low oxygen levels and strong nutrient-rich upwelling that facilitated the deposition of biogenic siliceous material like diatomite, which was later altered to chert. The presence of detrital zircon spectra dominated by Sierra Nevada sources indicates sediment transport from the Sierran arc across the forearc basin during the Monterey Group deposition.

The Claremont Formation marks a period of subsidence in the forearc basin, which was influenced by the transition from a convergent to a transform boundary. The erosional unconformity at the top of this unit represents a tectonic shift tied to the uplift of the East Bay block during the onset of the San Andreas transform regime.



Figure 4: *Field photo showing a well-exposed section of the Claremont Formation and location. This site highlights the interbedded chert and shale layers characteristic of a deep marine depositional environment.*

Orinda Formation (Late Miocene)

Overlying the Claremont Formation is the Orinda Formation, characterized by fluvial sandstones, conglomerates, and red siltstones. These deposits represent terrestrial environments, including river channels and floodplains. The clast composition reflects a transition from Sierra Nevada-derived material to local sources, including Franciscan Complex detritus, indicating uplift and erosion of the Coast Ranges during the Mendocino Triple Junction's northward migration.

The Orinda Formation records basin subsidence and the development of local depositional centers along active faults, such as the Hayward Fault. The interbedded lithologies suggest episodic fluvial sedimentation, with changes in sediment provenance reflecting the progressive uplift and erosion of the surrounding highlands.

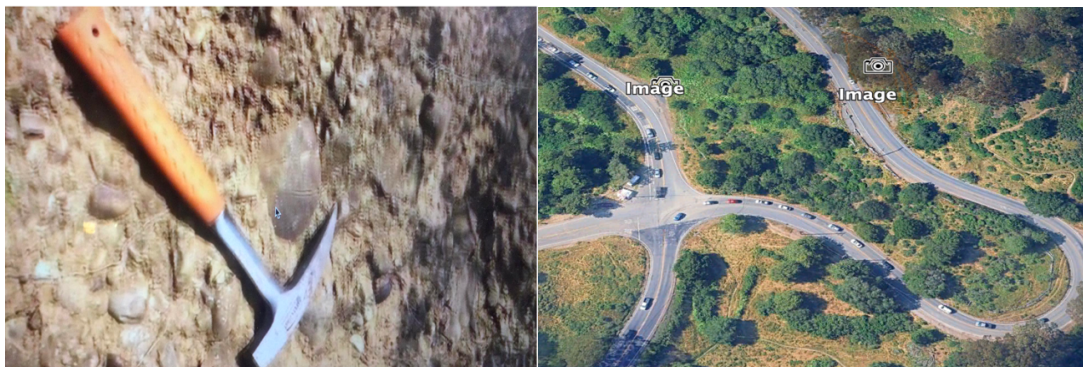


Figure 5: *Outcrop of the Orinda Formation and its location, illustrating fluvial conglomerates and sandstones deposited in a terrestrial river channel environment.*

Lower Moraga (Late Miocene)

The Lower Moraga is the basal volcanic unit of the Moraga Formation, primarily composed of basaltic lava flows. It reflects the early phase of volcanic activity in the Berkeley Hills. The unit includes several subunits, each representing distinct depositional and volcanic processes:

Subunits :

Lower Moraga Tuff: This subunit consists of fine-grained volcanic tuff, often red to white, formed from lithified volcanic ash deposits. The tuff displays bedding indicative of pyroclastic flow deposition, with occasional accretionary lapilli observed, suggesting

subaerial explosive eruptions.

Moraga Lapillistone: Composed primarily of lapilli-sized pyroclasts, this unit exhibits a coarse texture with clasts set in a fine ash matrix. The lapillistone often shows sorting, with coarser fragments at the base of beds, indicating deposition from pyroclastic surges.

Moraga Volcaniclastic Breccia: This unit is characterized by angular to subangular volcanic clasts in a finer-grained matrix. The breccia is poorly sorted, reflecting deposition during explosive eruptions and subsequent gravitational collapse of unstable volcanic edifices.



Figure 6: *Outcrop of the Lower Moraga Formation and location, showcasing the dark basaltic lava flows indicative of early effusive volcanic activity.*



Figure 7: *Exposure of the Lower Moraga Tuff and location, displaying fine-grained volcanic ash deposits with bedding indicative of pyroclastic flow deposition.*



Figure 8: *Lower Moraga Lapillistone outcrop, composed of coarse lapilli fragments deposited during pyroclastic surges.*

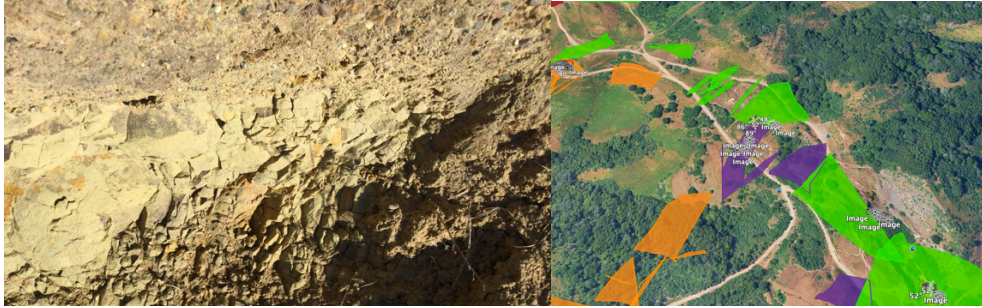


Figure 9: *Lower Moraga Volcaniclastic Breccia observed, featuring angular volcanic clasts within a fine-grained matrix, indicative of explosive volcanic activity.*

Moraga Tuff

The Moraga Tuff is a distinctive reddish-white, fine-grained volcanic tuff unit that separates the Lower Moraga from the overlying Moraga Sediment. The fine-grained nature of the tuff suggests deposition from widespread ashfall events. Thin bedding and cross-lamination are observed, indicative of transport and reworking by wind and water after initial deposition.



Figure 10: *Exposure of the Moraga Tuff.*

Moraga Sediment

The Moraga Sediment comprises interbedded sandstones and siltstones with occasional pebbly conglomerates. These sediments are often quartz-rich, with feldspar and volcanic detritus. Crossbedding and ripple marks are commonly observed, suggesting deposition in fluvial or shallow lacustrine environments. These deposits likely represent intervals of reduced volcanic activity, allowing sedimentation to dominate.

Upper Moraga (Late Miocene)

The Upper Moraga marks the final phase of volcanic activity in the Moraga Formation. It is composed of dark gray, fine-grained basalt flows with "onionskin" weathering textures. These basalts are rich in feldspar and olivine, indicative of a more evolved magma source. Flow banding is occasionally observed, and columnar jointing suggests cooling and solidification under subaerial conditions.

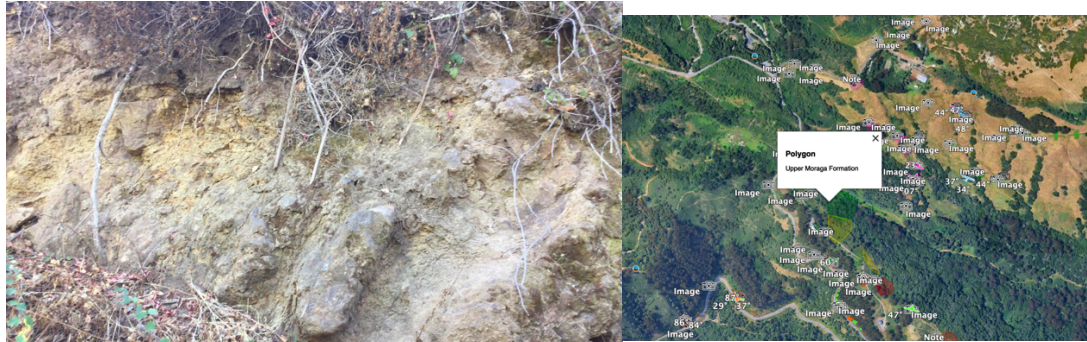


Figure 11: *Upper Moraga Formation outcrop, highlighting basalt flows with "onion-skin" weathering indicative of evolved magma sources.*

Siesta Formation (Late Miocene)

The Siesta Formation is composed of lacustrine claystones, siltstones, sandstones, and occasional limestones. These fine-grained sediments indicate deposition in quiet, low-energy lake environments, likely within an internally drained basin. The presence of bentonite layers and fossiliferous horizons supports an interpretation of episodic volcanic ash falls and stable lake conditions, respectively.

This unit marks a transition to episodic lacustrine deposition associated with continued basin subsidence and fault-controlled isolation of the depositional environments.



Figure 12: *Siesta Formation exposed, characterized by lacustrine claystones and siltstones indicating deposition in low-energy lake environments.*

Bald Peak Formation (Late Miocene)

Capping the sequence is the Bald Peak Basalt, a series of massive basalt flows dated to ~8.4 Ma. These terrestrial lava flows represent the final phase of volcanic activity in the region, coinciding with the waning effects of slab-gap volcanism. Pillow structures observed in some flows suggest subaqueous emplacement, potentially in lacustrine environments during periods of higher water levels.

The Bald Peak Formation signifies the cessation of active volcanism in the Berkeley Hills, with

subsequent tectonic processes shaping the region into its modern configuration. This transition is evidenced by the erosional truncation of these flows and their deformation within the Siesta syncline.



Figure 13: *Massive basalt flows of the Bald Peak Basalt Formation*

MAGMATIC AND VOLCANIC ACTIVITY – A FOCUS ON SIBLEY

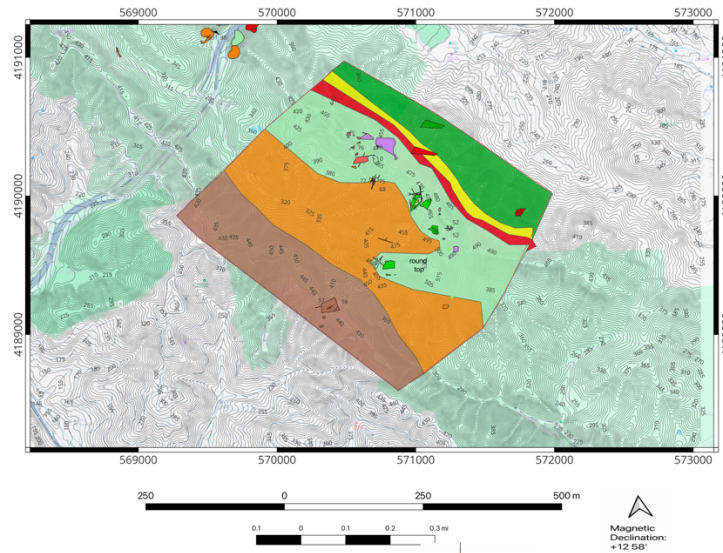


Figure 14: Geological Formation of Sibley Volcanic Preserve

The Sibley Volcanic Regional Preserve, situated in the Berkeley Hills, offers a remarkable window into the volcanic history of the East Bay region. The area's geology is dominated by the Moraga Formation, a sequence of volcanic rocks that erupted during the late Miocene epoch, approximately 10.2 to 9 million years ago.

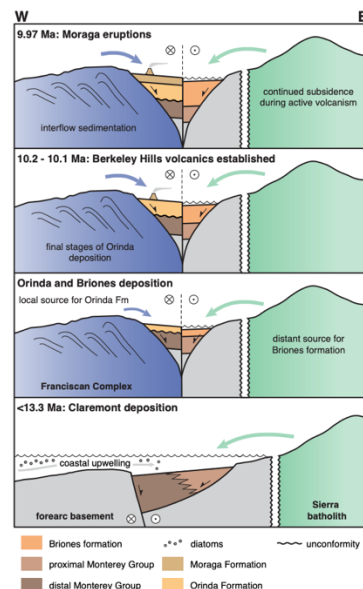


Figure 6. Simplified schematic illustration of sediment provenance and tectonic history of basin development associated with Miocene strata in the Berkeley Hills region. Basin geometry is characteristic of modeled transtensional strike-slip motion (Wu et al., 2009). The breaks between the forearc basement and Sierra batholith represent a large amount of distance. We interpret Claremont formation (Monterey Group) deposition to have occurred during initial regional transtension associated with the arrival of the Mendocino triple junction (Atwater and Stock, 1998). Zircon within the marine Claremont formation were dominantly sourced from the Sierra batholith. Uplift was followed by renewed subsidence leading to deposition of terrestrial sediments of the Orinda Formation, which were sourced from uplifted Franciscan Complex lithologies. Broadly time-equivalent marine sandstones of the Briones formation maintain the same Sierran sourced provenance as the Claremont formation. Slab-gap volcanism then initiated during ongoing subsidence leading to the eruption of the Moraga Formation volcanics during terminal Orinda Formation deposition.

Figure 15: Simplified schematic illustrating the sediment provenance, tectonic history, and basin development associated with Miocene strata in the Berkeley Hills region.

This cross-section (Fig13) shows the transition from Claremont Formation deposition during

early regional transtension (<13.3 Ma) to Orinda Formation sedimentation (local and distal sources), and the establishment of volcanic activity during the Moraga Formation eruptions (9.97 Ma). The schematic highlights basin subsidence, sediment transport dynamics, and volcanic processes linked to the Mendocino Triple Junction and slab-gap volcanism.

Geological Evolution and Formation Processes

The volcanic activity in the Sibley area is closely linked to tectonic movements along the San Andreas Fault system. The passage of the Mendocino Triple Junction—a point where three tectonic plates meet—resulted in the formation of a slab gap beneath the region. This gap allowed hot asthenospheric mantle material to upwell, leading to decompression melting and the generation of magma. The resulting volcanic activity produced a complex field of basaltic lava flows and associated pyroclastic deposits (Flower & Strong, 1993; Dickinson, 1981).

The Moraga Formation within the Sibley Preserve comprises multiple volcanic units, each representing distinct eruptive phases:

Lower Moraga includes basaltic lava flows interbedded with pyroclastic deposits such as tuffs and breccias, indicating alternating explosive and effusive eruptions.

Moraga Tuff is a prominent layer of volcanic ash, suggesting a significant explosive event that dispersed ash over a wide area. Upper Moraga Volcanics is characterized by additional basaltic flows, marking a return to effusive volcanic activity.

Interspersed between these volcanic layers are sedimentary deposits, including fluvial and lacustrine sediments, which accumulated during periods of volcanic quiescence. These interbeds provide evidence of changing environmental conditions and pauses in eruptive activity (Creely, 1982).

Eruptive Units and Source Areas

Detailed mapping and field studies have identified several key features

Lava Flows: The basaltic flows exhibit characteristics such as columnar jointing and vesicular textures, indicative of cooling and degassing processes. The presence of autoclastic basaltic breccia—formed from the fragmentation of lava during flow—suggests dynamic eruptive conditions.

Volcanic Vents: The Round Top area is interpreted as a central vent complex, now tilted and exposed due to tectonic forces. This vent was likely the source for many of the lava flows observed in the preserve.

Dikes and Sills: Intrusive features such as dikes and sills are present, representing pathways through which magma ascended to the surface. These structures provide insights into the subsurface plumbing system of the ancient volcanic field (Sarna-Wojcicki et al., 2011).

The volcanic deposits in Sibley are part of a larger, compositionally diverse volcanic field that has been dismembered by faulting associated with the San Andreas Fault system. This tectonic activity has transported and deformed volcanic units, complicating the reconstruction of the original volcanic landscape (Graham, 1984; Atwater & Stock, 1998).

Volcanism in the California Coastal Ranges

The volcanic activity observed in the Sibley area is a manifestation of broader geodynamic processes affecting the California Coastal Ranges. The interaction between the Pacific and North American plates, particularly the transition from subduction to transform tectonics, has played a crucial role in generating magma. The slab window created by the northward migration of the Mendocino Triple Junction facilitated mantle upwelling and subsequent volcanism.

STRUCTURAL INTERPRETATION

The Berkeley Hills region exhibits a complex structural framework characterized by folds, faults, and unconformities resulting from tectonic processes along the active transform margin of the San Andreas Fault system. Detailed structural analysis, including field measurements and stereonet projections, provides insights into the deformation history and current seismicity of the area. (Jones, 1991; Graham, 1984).

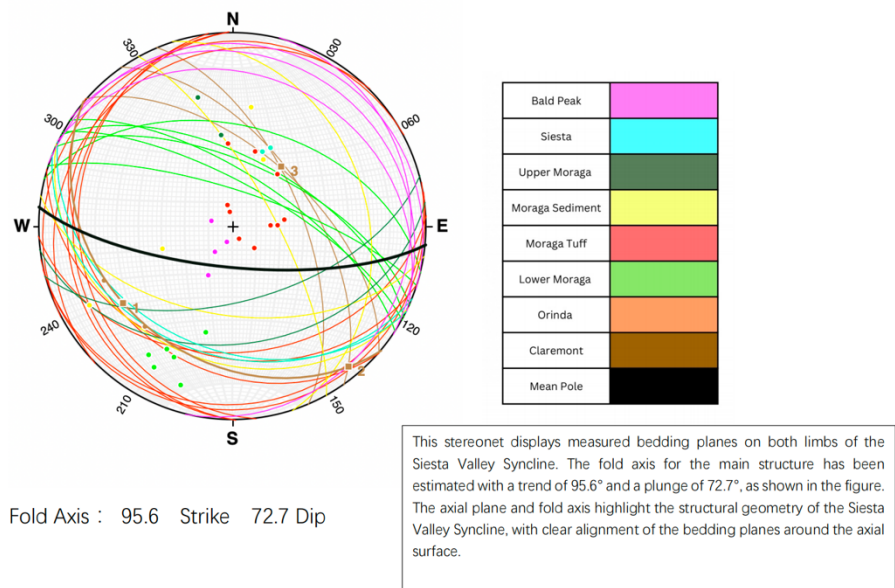


Figure 16: Stereonet diagram showing measured bedding planes on both limbs of the Siesta Valley Syncline. This analysis underscores compressional forces-oriented NW-SE during deformation.

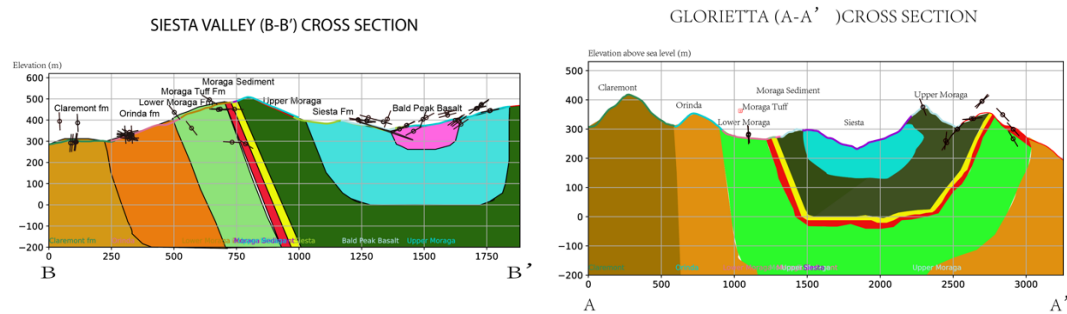


Figure 17: Cross-sections of the Berkeley Hills illustrating the stratigraphic and structural framework along two transects: Siesta Valley (B-B') and Glorietta (A-A').

Folds

The region features prominent folding, with the Siesta Valley syncline being a significant structure:

This syncline trends approximately NE-SW, with an axial plane orientation starting at $\sim 030^\circ$, as determined using the axial plane finder tool in stereonet analysis. The fold axis plunges gently to the southwest, suggesting compressional forces oriented NW-SE during deformation.

Faults

Faulting in the Berkeley Hills is dominated by the Hayward Fault Zone:

Hayward Fault: A major right-lateral strike-slip fault running along the base of the Berkeley Hills, separating the hills from the adjacent flatlands. It exhibits active creep at a rate of approximately 0.1 inches per year near the University of California, Berkeley campus (Dickinson, 1981).

Wildcat Fault: A secondary fault within the San Andreas system, traversing the canyon near the Botanical Garden. Its activity contributes to the complex deformation patterns observed in the area (Gerasimov et al., 2023).

Age and Causes of Deformation

The deformation in the Berkeley Hills is primarily attributed to the following tectonic events:

Miocene Epoch: The transition from subduction to transform tectonics along the evolving San Andreas Fault system led to significant crustal deformation, including folding and faulting. The development of the Siesta Valley syncline and associated structures likely occurred during this period.

Pliocene to Present: Ongoing right-lateral strike-slip motion along the Hayward Fault continues to deform the region, contributing to seismic activity and landscape evolution.

Structural Measurements and Stereonet Analysis (Fig 14)

Field measurements of bedding planes within the Siesta Valley syncline were plotted using stereonet software. The axial plane finder tool identified an axial plane striking NE-SW ($\sim 030^\circ$) with a gentle southwestward plunge. This analysis corroborates field observations of fold geometry and orientation (Flower & Strong, 1993).

CONCLUSIONS

The Berkeley Hills' geological record, encompassing formations such as the Claremont, Orinda, and Moraga, encapsulates a dynamic history of tectonic transitions from a convergent to a transform margin. This evolution, driven by interactions between the Pacific and North American Plates, led to significant basin development, volcanic activity, and structural deformation. Detailed studies, including those by Gerasimov et al. (2024), have provided precise chronostratigraphic frameworks that enhance our understanding of these processes. The integration of field observations with advanced geochronological techniques has been instrumental in reconstructing the complex geological history of the Berkeley Hills, offering insights into the broader tectonic evolution of California's Coast Ranges.

REFERENCES CITED

Creely, S., Savage, D. E., & Ogle, B. A., 1982, Stratigraphy of Upper Tertiary Nonmarine Rocks of Central Contra Costa Basin, California, In: Ingersoll, R.V., and Woodburne, M.O. (eds.), *Cenozoic Nonmarine Deposits of California and Arizona: Pacific Section*, SEPM, pp. 122.

Dickinson, W. R., 1981, Plate Tectonics and the Continental Margin of California, *Geological Society of America Bulletin*, 92, 7–22.

Flower, B. P., & Kennett, J. P., 1993, Relations between Monterey Formation deposition and middle Miocene global cooling: Naples Beach section, California, *Geological Society of America Bulletin*, 105(10), 1265–1273.

Fox, K. F., Fleck, R. J., Curtis, G. H., & Meyer, C. E., 1985, Implications of the Northwestwardly Younger Age of the Volcanic Rocks of West-Central California, *Geological Society of America Bulletin*, 96, 647–654.

Gerasimov, S. H., Hodgins, E. B., Crowley, J. L., & Swanson-Hysell, N. L., 2023, Chronostratigraphy of Miocene Strata in the Berkeley Hills and the Arrival of the San Andreas Transform Boundary, *Geological Society of America Bulletin*, 135(4–5), 763–784.

Graham, S. A., McCloy, C., Hitzman, M., Ward, R., & Turner, R., 1984, Basin Evolution During Change from Convergent to Transform Continental Margin in Central California, *American Association of Petroleum Geologists Bulletin*, 68(3), 233–249.

Jones, D. L., & Curtis, G. H., 1991, Guide to the Geology of the Berkeley Hills, Central Coast Ranges, California, *California Geological Society Annual Report*, 14, 25–40.

Sarna-Wojcicki, A. M., Deino, A. L., Fleck, R. J., et al., 2011, Age, Composition, and Areal Distribution of the Pliocene Lawlor Tuff, and Three Younger Pliocene Tuffs, California and Nevada, *Geosphere*, 7(3), 599–628.

Fox, K. F., & Dickinson, W. R., 1976, Neogene Volcanism and Tectonism in the Coastal Ranges, *Geological Society of America Memoir*, 134, 45–58.

Edwards, S. C., & Erskine, M. C., 2005, Robert Sibley Volcanic Regional Preserve in the Berkeley Hills, Northern California Geological Society Field Trip Guidebook, 5(2), pp. 1–16.

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